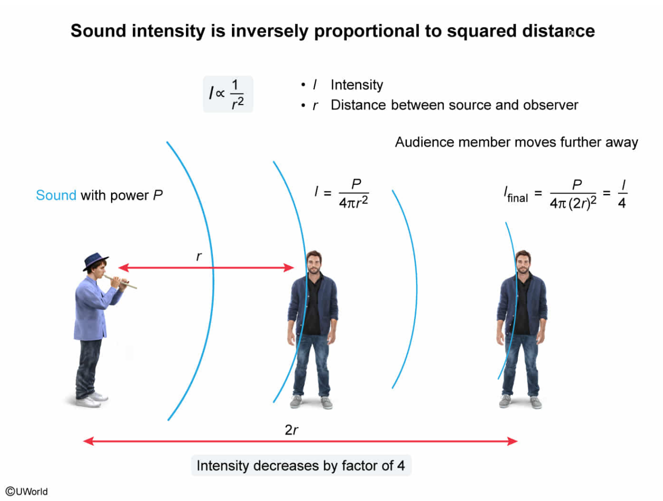
An audience member experiences sound of intensity *I* as an orchestra member plays a flute. If the audience member moves farther back so that the distance from the flute player is doubled, then *I* would:

- A. decrease by a factor of 4. [63%]
- B. decrease by a factor of 2. [29%]
- C. remain the same. [6%]
- D. increase by a factor of 2. [1%]

Correct
62% answered correctly

Explanation:



The loudness of a sound wave is measured in terms of the wave's intensity. The intensity *I* of a sound wave is equal to the energy *E* carried by the sound wave per unit area *A* per unit time *t* :

$$I = \frac{E}{A \cdot t}$$

Because sound waves move in three dimensions, the intensity can be expressed in terms of the **power** *P* per *A*, where *A* is the surface area of a sphere with radius *r* :

$$A = 4\pi r^2$$

Hence:

$$I = \frac{P}{A} = \frac{P}{4\pi r^2}$$

Consequently, intensity is directly proportional to *P* and inversely proportional to the *square* of the distance *r* from the sound source:

$$I \propto \frac{1}{r^2}$$

In this question, the audience member initially experiences an intensity I_{inital} at a distance *r* from the source:

$$I_{\text{initial}} = \frac{P}{4\pi r^2} = I$$

When the audience member moves farther away, the power of the sound remains the same but the distance is doubled. Hence, the final intensity equals:

$$I_{\text{final}} = \frac{P}{4\pi (2r)^2}$$

$$I_{\text{final}} = \frac{1}{4} \cdot \left(\frac{P}{4\pi r^2} \right) = \frac{1}{4} I$$

Therefore, the intensity of the sound decreases by a factor of 4.

(Choice B) Intensity would decrease by a factor of 2 if it was inversely proportional to *r*, but it is inversely proportional to r^2.

(Choice C) The energy of the sound is unchanged, but intensity also depends on the distance from the source.

(Choice D) Intensity can only decrease as the distance between source and observer increases.

Educational objective:
Sound intensity equals the rate at which energy is delivered per unit surface area, and is inversely proportional to the square of the distance from the source.

Time spent:0 sec
Subject:
Physics

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System:
Light and Sound